

**Course: Fundamentals of Generative AI and Large Language
Models: Theory and Practice**

**Week 2: Autoencoders: The Representation
Learning Model**

Week 2 Outline

Autoencoders

- ✓ Introduction to Autoencoders
- ✓ Architecture of Autoencoders
- ✓ Undercomplete vs Overcomplete AEs
- ✓ Training objective: Reconstruction Loss (MSE, Binary Cross-Entropy)
- ✓ Types of Autoencoders (Simple, Deep, CNN-based)
- ✓ Regularization Techniques in AE: Denoising, Sparse, Contractive
- ✓ Understanding Autoencoders through a Numerical Example
- ✓ Limitations of Standard Autoencoders → why VAEs are needed

Hands-on Implementation of Types of Autoencoders

Today's Session Overview

✓ Autoencoders

- ✓ Introduction
- ✓ Architecture of Autoencoders
- ✓ Undercomplete vs Overcomplete AEs

From Supervised Learning to Autoencoders

Most traditional neural network architectures such as ANNs, CNNs, and RNNs are primarily used in supervised learning, where:

Input X + Label Y



Model learns $X \rightarrow Y$

F	G	H	I	J	K	L	M	N	O
		Sepal Length	Sepal Width	Petal Length	Petal Width	Class			
Xtrain		5.1	3.5	1.4	0.2	setosa	Y-Train		
		4.9	3	1.4	0.2	setosa			
		7	3.2	4.7	1.4	versicolor			
		6.4	3.2	4.5	1.5	versicolor			
		6.9	3.1	4.9	1.5	versicolor			
		6.6	3	4.4	1.4	versicolor			
		4.7	3.2	1.3	0.2	setosa			
		5.1	3.5	1.4	0.3	setosa			
		5.2	3.5	1.5	0.2	setosa			
		6.3	2.9	5.6	1.8	virginica			
		6.5	3	5.8	2.2	virginica			
		7.6	3	6.6	2.1	virginica			
		5.2	3.4	1.4	0.2	setosa			
		4.7	3.2	1.6	0.2	setosa			
		4.8	3.1	1.6	0.2	setosa			
XTest		5.4	3.4	1.5	0.4	setosa	YTest		
		6.8	2.8	4.8	1.4	versicolor			
		6.7	3	5	1.7	versicolor		setosa	
		6	2.9	4.5	1.5	versicolor		versicolor	
		5.8	2.7	5.1	1.9	virginica		virginica	
		7.1	3	5.9	2.1	virginica		virginica	
Ypred		4.9	2.5	4.5	1.7	virginica		setosa	
		7.3	2.9	6.3	1.8	virginica		virginica	
		6.7	2.5	5.8	1.8	virginica		versicolor	

Training dataset

But what if labels are not available?

Age (years)	Height (cm)	Weight (kg)
6	115	20
8	125	25
10	135	30
12	145	36
14	155	45
16	165	55
18	170	62
20	172	66
22	174	69
25	175	72

Unsupervised learning:
where the model learns directly from unlabeled data.

In unsupervised learning:

Only Input X



Model learns hidden patterns, structures, and correlations in X.

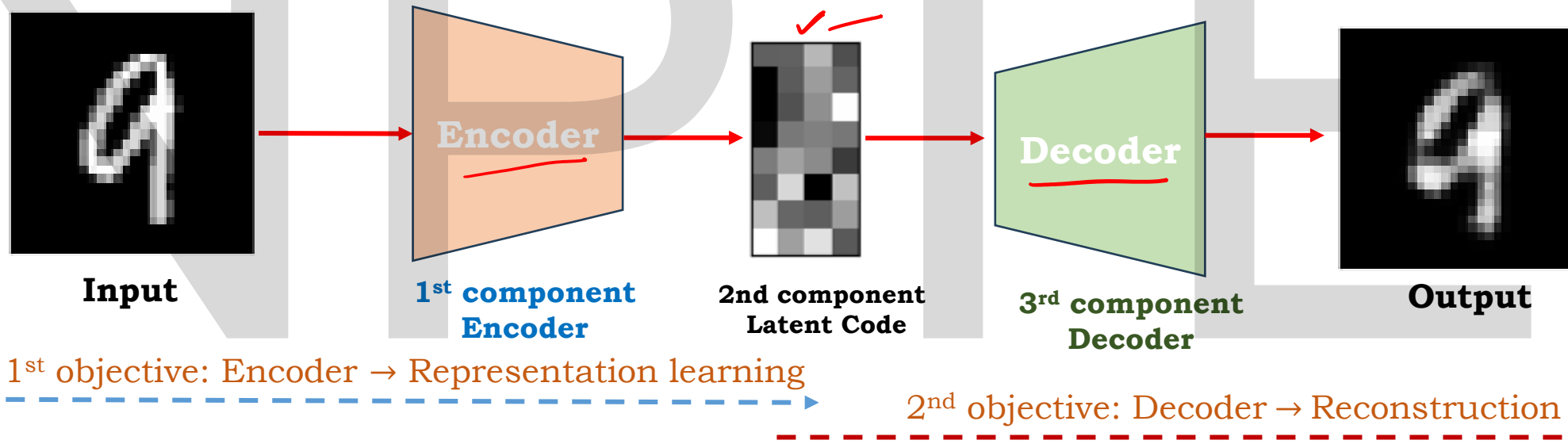
While unsupervised learning broadly focuses on discovering patterns and structure in unlabelled data, a special category of unsupervised deep learning models aims to **learn meaningful representations of the data itself.**

Autoencoders are such special category of unsupervised deep learning model that learn meaningful representations by reconstructing their own inputs.

Architecture of Autoencoder

Autoencoders have two major objectives:

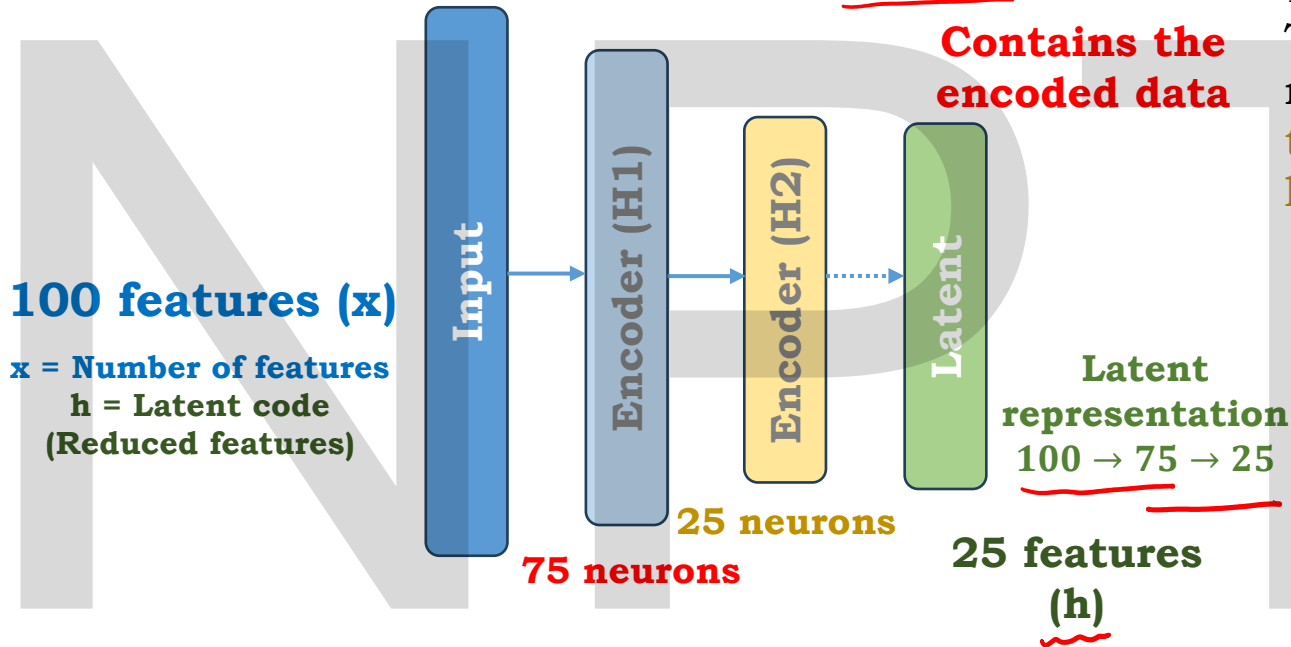
- 1. Representation learning:** The encoder learns a compact latent representation of the input, enabling unsupervised feature learning.
- 2. Reconstruction learning:** The decoder reconstructs the original input from the latent representation.



Input is passed to the encoder; encoder learns meaningful features or representations of the data. The output of the encoder is the smaller internal representation (**Latent code**), and this representation is passed to the decoder which tries to reconstruct the original input. The special idea in autoencoders is that the input itself is used as the target.

Encoder Objective: Compression and Representation Learning

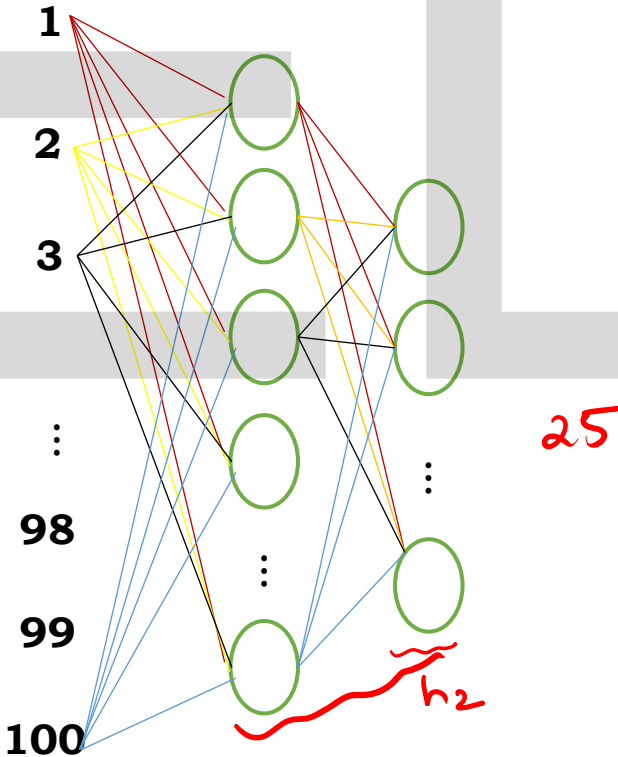
These 75 values are then passed to the next hidden layer with 25 neurons. Again, each neuron combines information from the previous layer to produce a single output value. Hence, we get 25 values. 75 → 25



Input: 100 features
Hidden layer: 75 neurons

Suppose the input has 100 features. These 100 features are passed to the first hidden layer with 75 neurons. Since this is a fully connected encoder, each neuron receives information from all 100 input features and produces one output value. Therefore, the 75 neurons produce 75 values. $100 \rightarrow 75$

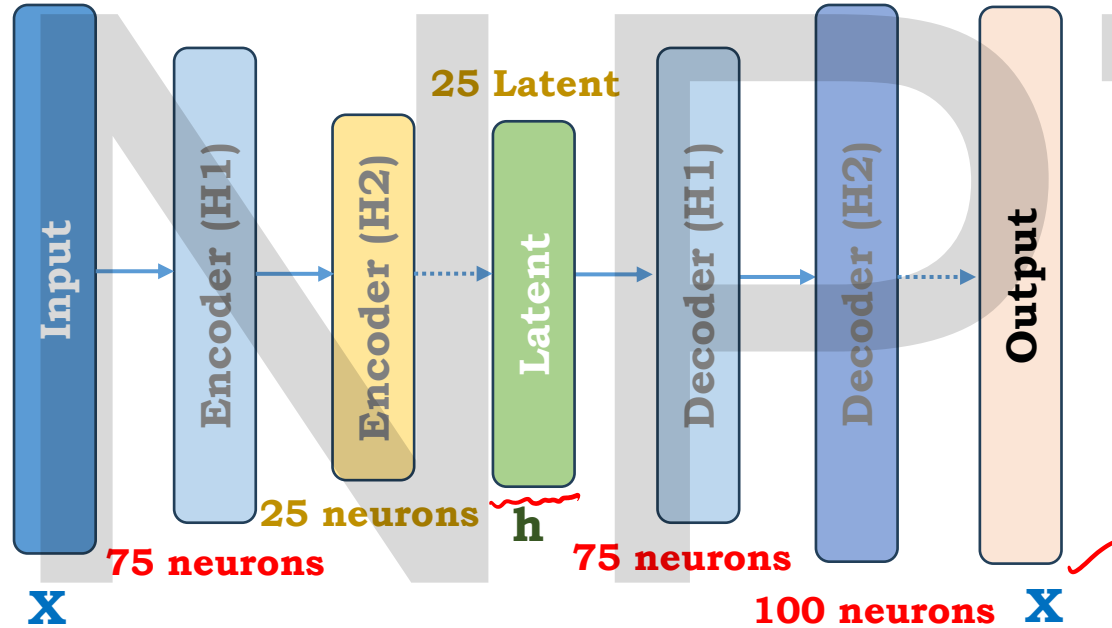
Encoder maps the input data → lower-dimensional latent representation by progressively reducing the number of hidden units. The encoder reduces the input dimension from $100 \rightarrow 75 \rightarrow 25$. This reduction forces the network to preserve only the most important information from the input. Therefore, the first objective of the autoencoder, representation learning, is achieved.



Decoder objective: Reconstructing the Input from the Latent Code

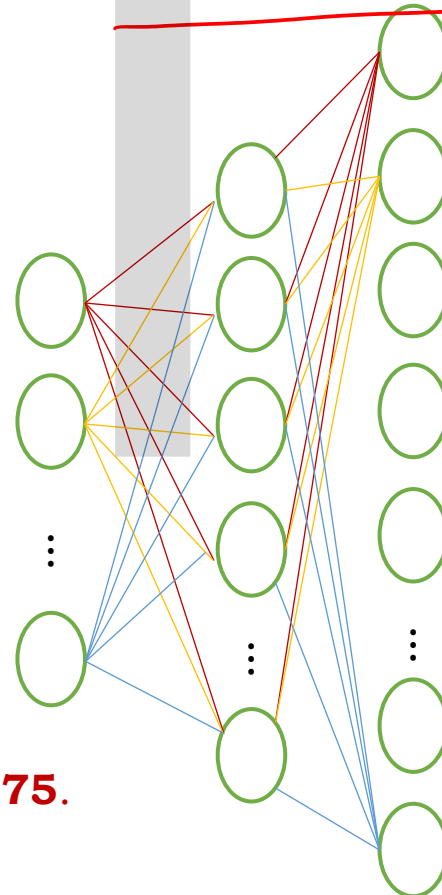
Then these 75 values are passed to the output layer with 100 neurons. Since the original input had 100 features, the output layer also produces 100 values. These 100 values form the reconstructed output.

The decoder takes the **latent representation** as its input



The decoder does not simply recover the original input by copying it. It reconstructs using only the compressed information available in the latent code.

If the latent code has captured the important structure of the input, the reconstruction will be close to the original input. **Therefore, the second objective of the autoencoder, reconstruction learning, is achieved.**

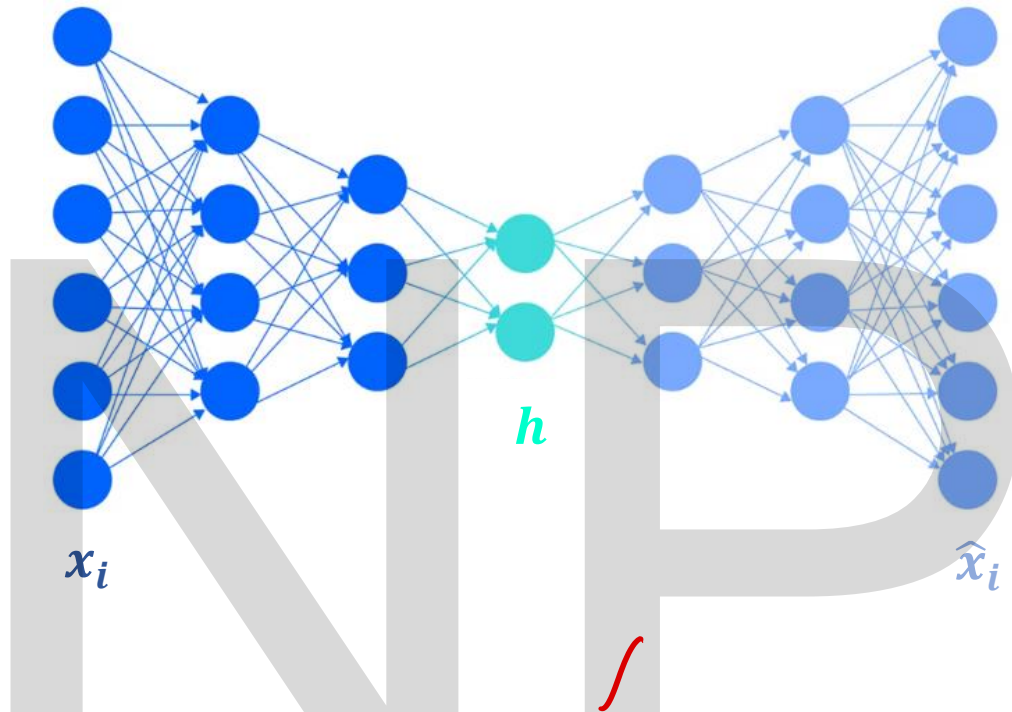


These 25 values are passed to a hidden layer with 75 neurons. Each neuron combines information from the latent code to produce a single output value.

Therefore, this layer produces 75 values.

Hence, the decoder performs an **expansion from 25 → 75**.

Autoencoder Representation: Undercomplete vs. Overcomplete



Case 1: $\dim(h) < \dim(x_i)$

This is called an undercomplete autoencoder.

Since the hidden representation has fewer dimensions than the input, the model is forced to learn the most important features needed for reconstruction.

If we are still able to reconstruct $\hat{x}_i$ perfectly from h , then h is called loss-free encoding of x_i .

It captured all the important characteristics of x_i .

An autoencoder is a feed-forward neural network that learns to reconstruct its input.

Encoder: Maps the input x_i to a hidden representation h .

Decoder: Maps the hidden representation h back to a reconstructed output $\hat{x}_i$.

The main aim here is to learn the most important characteristics of the input data x_i

$$x_i \rightarrow h \rightarrow \hat{x}_i$$

Then h has captured everything that is required to reconstruct $\hat{x}_i$ from x_i

Case 2: $\dim(h) \geq \dim(x_i)$

Since the hidden representation is equal to or larger than the input dimension, In such a case the autoencoder could learn a trivial encoding by copying x_i into h and then copying h into $\hat{x}_i$. This is called overcomplete autoencoder.

Example: $4 \rightarrow 6 \rightarrow 4$

Next session

Training objective: Reconstruction Loss Function (MSE, Binary Cross-Entropy)

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Thank you